

1. a REF (b) RREF (c) Neither (d) Neither

2. (a) $\begin{bmatrix} 3 & 0 & 9 \\ 0 & 1 & 3 \\ 1 & 1 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 6 \\ 9 \end{bmatrix}$ (b) Augmented $\begin{bmatrix} 3 & 0 & 9 & 12 \\ 0 & 1 & 3 & 6 \\ 1 & 1 & 6 & 9 \end{bmatrix}$

3. The solution set is infinite

4. $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 3 & 1 & 19 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & -5 & 10 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 7 \\ 0 & 1 & -2 \end{array} \right]$ so $\begin{bmatrix} 3 \\ 19 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 3 \end{bmatrix} - 2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

5. Strategy shows i,j entry of both sides are equal

$$i,j \text{ entry of } c(A+B) = c(a_{ij} + b_{ij}) = ca_{ij} + cb_{ij} =$$

$$i,j \text{ entry of } cA + i,j \text{ entry of } cB = i,j \text{ entry of } (cA + cB)$$

6. (a) rewrite as equations $x_1 = 4x_5 - 2x_3$

$$x_2 = 3x_5$$

$$x_3 = x_3$$

$$x_4 = -5x_5$$

$$x_5 = x_5$$

$$\therefore \vec{x} = s \begin{bmatrix} 4 \\ 3 \\ 0 \\ -5 \\ 1 \end{bmatrix} + t \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{with } s \text{ and } t \text{ any Real \#s}$$

(b) Solution set is a plane through origin spanned by

7. $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} -1 & 2 \\ 0 & -1 \end{bmatrix}$ are both invertible but

$$A+B = \begin{bmatrix} 0 & 4 \\ 0 & 0 \end{bmatrix} \text{ is not.}$$

8. (a) matrix (b) vector (c) scalar (d) gibberish (e) gibberish

9. (a) must have $ad-bc \neq 0$ $6 \cdot 3 - 2b \neq 0$ for all b except 9

(b) Augment by identity matrix, row reduce, if invertible, the identity becomes A^{-1}

$$\left[\begin{array}{ccc|ccc} -1 & -7 & -3 & 1 & 0 & 0 \\ 2 & 15 & 6 & 0 & 1 & 0 \\ 1 & 3 & 2 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 12 & 5 & 3 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & -9 & -4 & -1 \end{array} \right] \text{ so } A^{-1} = \begin{bmatrix} 12 & 5 & 3 \\ 2 & 1 & 0 \\ -9 & -4 & -1 \end{bmatrix}$$

10. (a) Not Lm and since there is no pivot in column 3,

(b) No since RREF would not have a pivot in Row 5

(c) No since A is not row equivalent to I

11. $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 3 \end{bmatrix}$

12. Equivalent to saying T is 1-1

(a) yes (b) No (c) No (d) No (e) Yes.

13. For all vectors $u \in V$ and all scalars c

(1) $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ (2) $T(c\vec{u}) = cT(\vec{u})$